A complete fifteen-variable consistency equation

Strictly positive integer unknowns; exact degree 157; a 66-operation evaluator.

The only unknowns are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

All other symbols in the equation are polynomial abbreviations:

$$\begin{gathered}
b = \beta J, \quad q = b + 1, \quad C = q - F - \alpha, \\
X = wq, \quad Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \\
\ell = EkY, \quad c = kY + \eta, \quad a = E + Y, \\
H = 4a + 3, \quad \Delta = a^2 + H, \quad D = X + ac + \gamma H, \\
R = [b(q - F) + \alpha](q^2 - 1) + (M + qN)J, \\
V = f^2(cT - R) - c, \quad S = i\Delta c^2, \\
P = \tau^2 - \ell(\ell + k), \quad Q = D^2 - \Delta c^2, \\
U = S^2(V^2 - y^2) + y^2, \quad W = k - hE - R, \\
Z = (K + w)C + (q - F) - zb, \quad G = \Delta f^2 - S^2.
\end{gathered}$$

The single equation is

$$\boxed{P\,Q\,U\,W\,Z\,G - \Delta = 0.}$$

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\begin{gathered}
\rho = 2^{47}, \quad B = \rho^{2963}, \quad \beta = B - 1, \\
g(m, x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1956}.
\end{gathered}$$

$$\begin{aligned}
M &= \beta - g(951, \rho) \\
&\quad - \rho^{960} g(26, \rho) - \rho^{1005}, \\
N &= \beta + 4g(917, \rho) + (2^{44} - 2)\Lambda \\
&\quad + \rho^{960} g(9, \rho) + \rho^{987} g(8, \rho) + \rho^{1005}, \\
K &= \rho^{18} + \rho^{1005} + \rho^{2972} \\
&\quad + \Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) = \sum_{i=0}^{941} [1 &+ 2^{10}(1 + w_i + 2^8 n_i \\
&+ 2^{17} p_i + 2^{25} s_i)] x^i \\
&+ 2^{10} x^{969}(x^9 + 2^{17}) g(8, 2x) \\
&+ 2^{18} x^{942}(x^{18} + 2^{17}) g(9, 2x).
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$\begin{gathered}
\theta(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\}, \\
r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j + 1$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, \theta(q, b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, \theta(q, b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $\mathrm{digit}(A, b, k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\begin{gathered}
\mathsf{mask} = \mathcal{A} \bmod 2^{636}, \\
\mathsf{targets} = \lfloor \mathcal{A}/2^{636} \rfloor \bmod 2^{1675}, \\
\mathsf{moves} = \lfloor \mathcal{A}/2^{2311} \rfloor, \\
\nu_j = \mathrm{digit}(\mathsf{mask}, 2, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k, \\
m_q = \mathrm{digit}(\mathsf{mask}, 2, 430 + q), \quad \delta_q = \sum_{k=0}^{q-1} m_k, \\
\omega_q = 14 + m_q(\mathrm{digit}(\mathsf{moves}, 16, \delta_q) - 14), \\
u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) \\
\cdot \mathrm{digit}(\mathsf{targets}, 216, j - \sigma_j), \\
v_j = \mathrm{digit}(\omega_q, 2, 2b), \quad d_j = \mathrm{digit}(\omega_q, 2, 2b + 1).
\end{gathered}$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Concatenate these thirteen lines, from top to bottom, to obtain its 781 decimal digits (there are no omitted digits):

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5860725087898678188912673742632546324025172298353855272995683176
4969949406333919315323872375782477466725463011528567718294055076
3006124739855037353655204691391414186758883217097375044976429465
8270944142085505376842559029102693041085562540684690710604079067
3724763129630582855384181944414973901733570469173085418871671744
2048170942517613524282407203789879020689779835988780400299936946
5274600673267211886931165288781352441481875991871486564767549564
1498891563641180775282772787179257559378852323557013970506009282
7980034336115111037194604982010168556898406861821168844529953973
3329097089832109899184569780832475009429376795970989537956914018
6560751892408082293940602808596063442474573309002707517217784697
0890324537666784074933609456239656992860826945102893145422737549
3204418786549
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The ordinary complemented-color, native and Pell arguments give unsolvability iff Con(ZFC). Native calibration and internal-ZFC formalization remain open in Lean. This page is a complete finite specification, not an unsolvability proof.